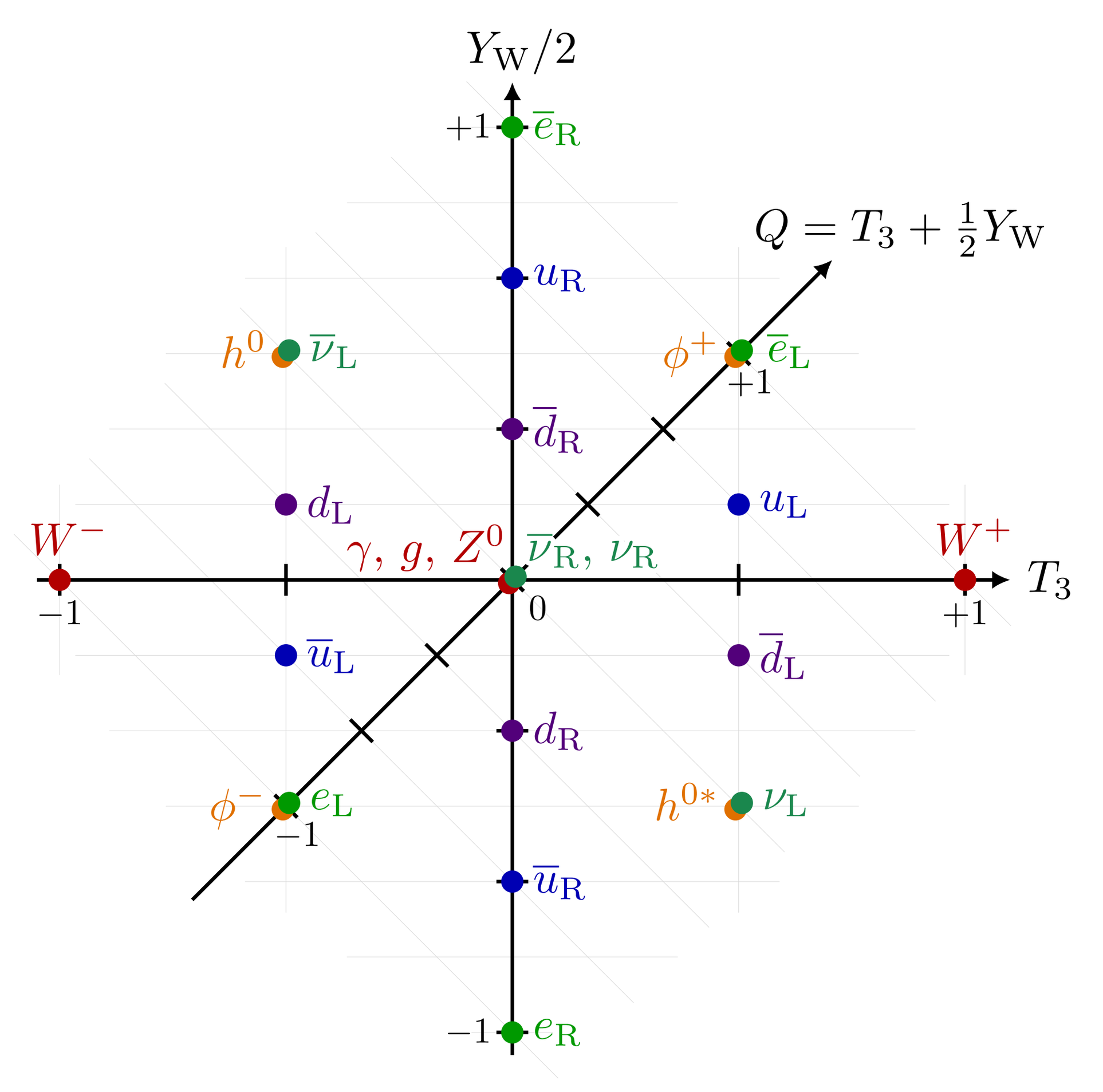
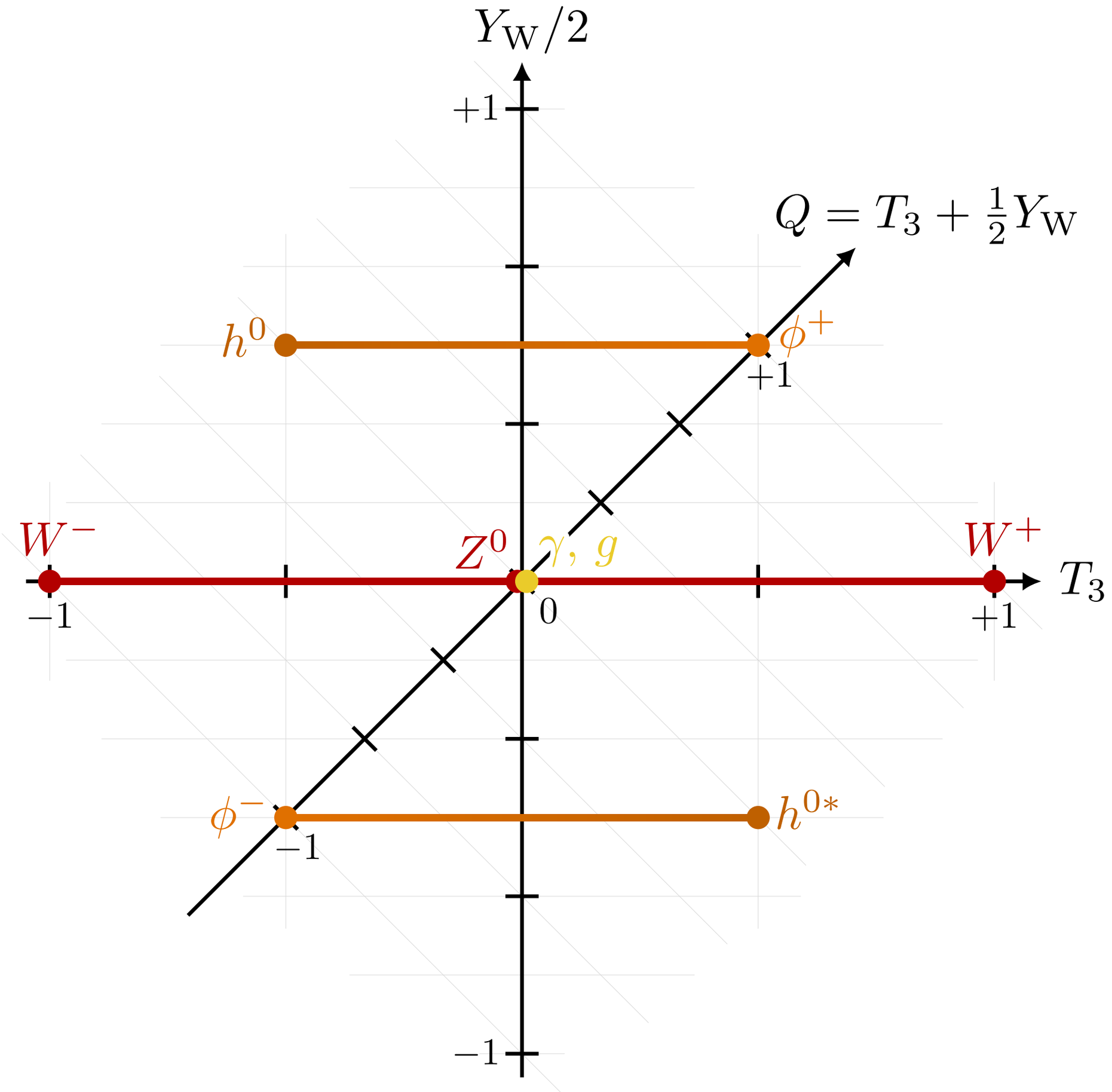
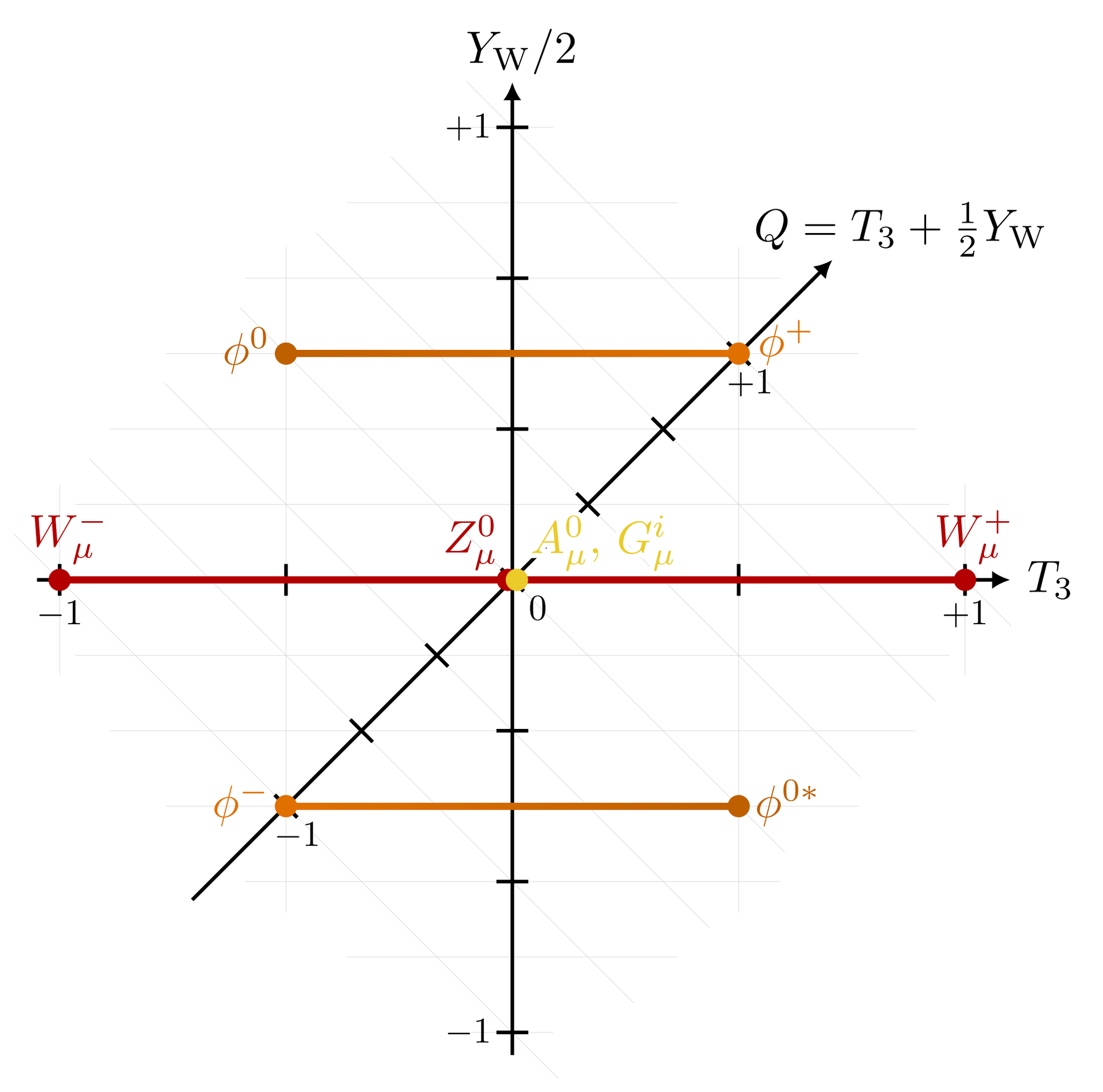
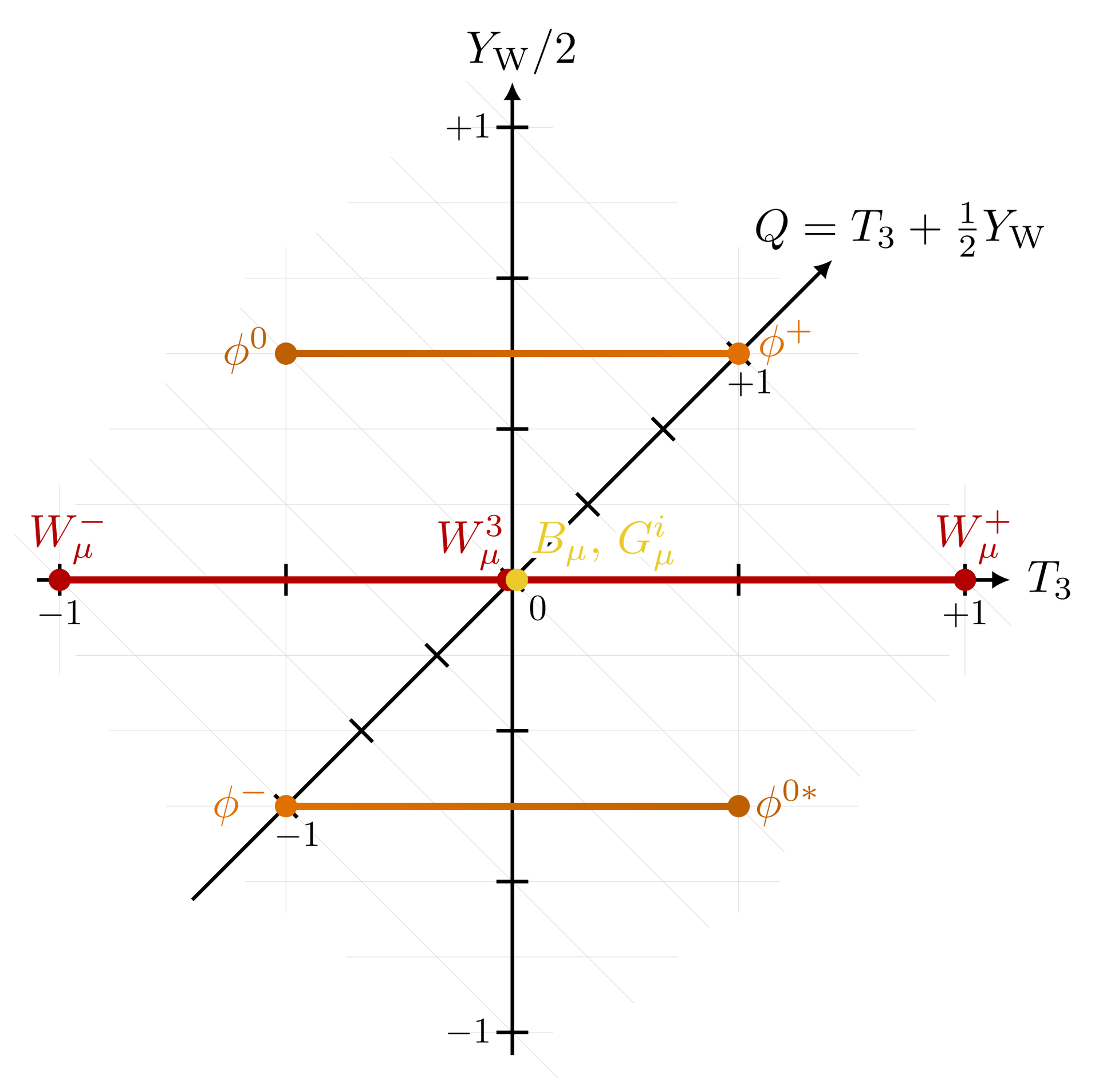
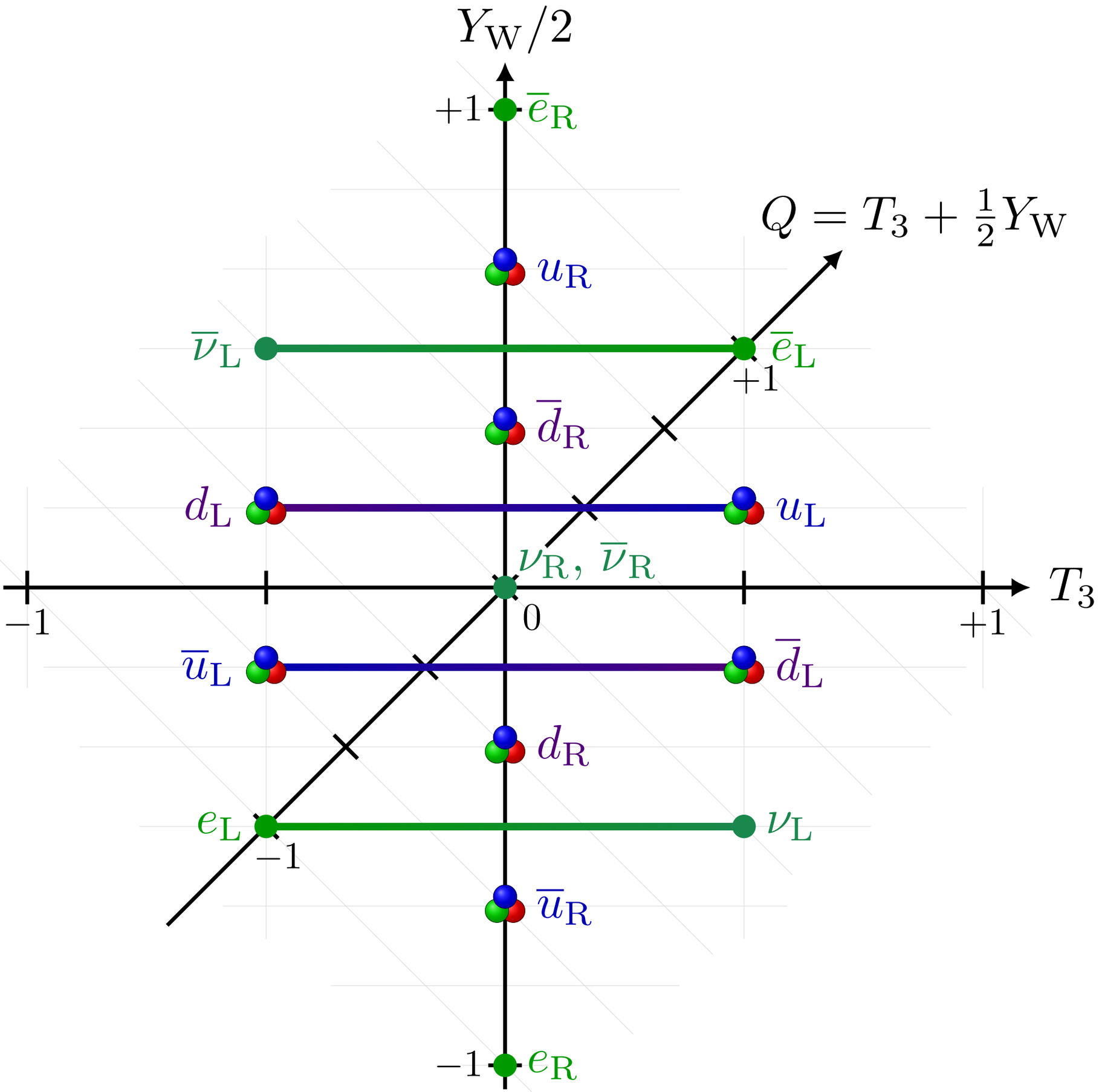
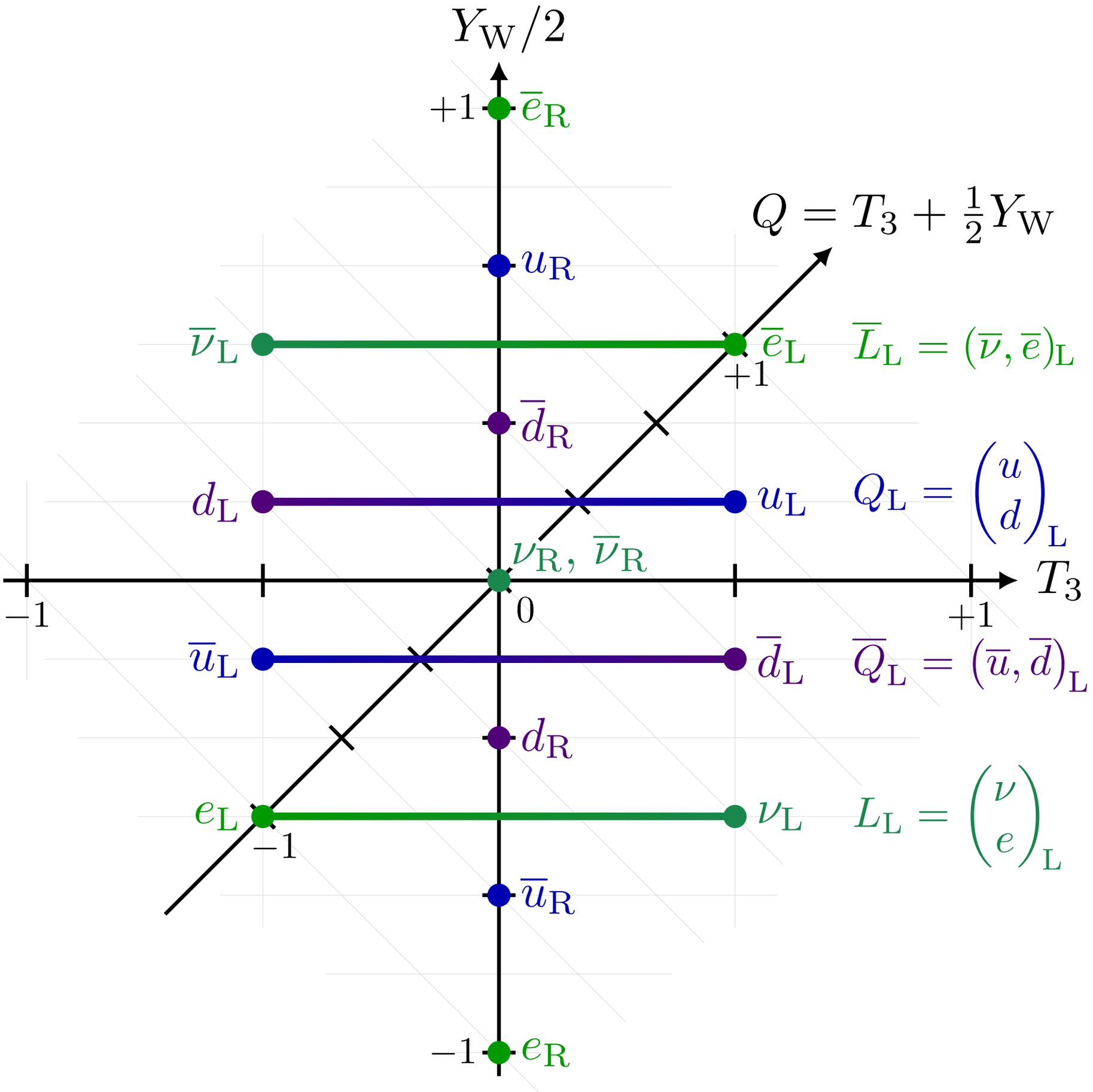

$Y_{\mathrm{W}}/2$
$Q = T_3 + \frac{1}{2}Y_{\mathrm{W}}$
T_3
$+1$
0
-1
$+1$
-1
$+1$
$\overline{e}_{\mathrm{R}}$
u_{R}
$\overline{d}_{\mathrm{R}}$
d_{R}
$\overline{u}_{\mathrm{R}}$
e_{R}
h^0
$\overline{\nu}_{\mathrm{L}}$
ϕ^+
$\overline{e}_{\mathrm{L}}$
d_{L}
u_{L}
W^-
$\gamma,\ g,\ Z^0$
$\overline{\nu}_{\mathrm{R}},\ \nu_{\mathrm{R}}$
W^+
$\overline{u}_{\mathrm{L}}$
$\overline{d}_{\mathrm{L}}$
ϕ^-
e_{L}
h^{0*}
ν_{L}

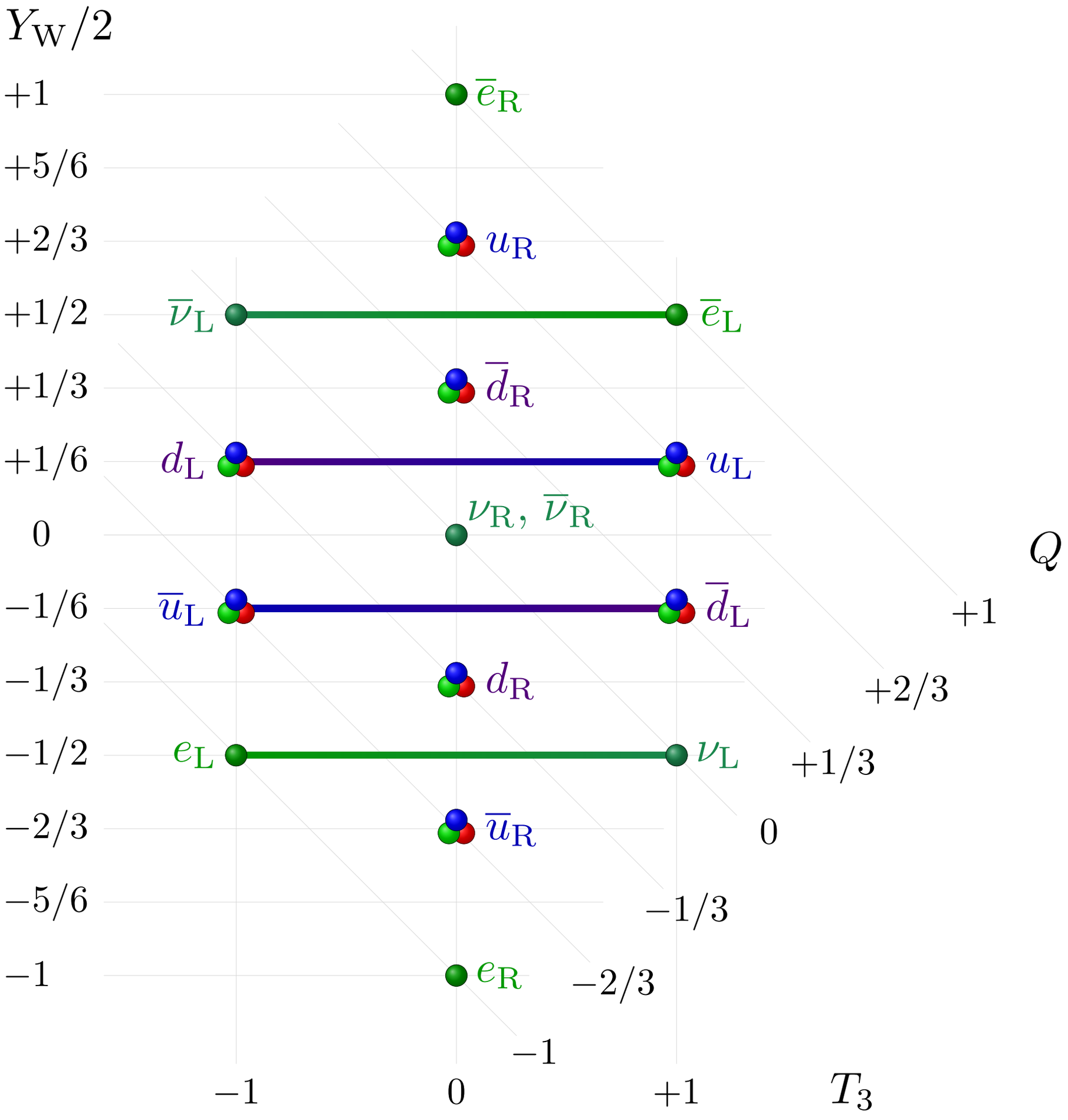

$Y_{\mathrm{W}}/2$
$+1$
$Q = T_3 + \frac{1}{2}Y_{\mathrm{W}}$
h^0
ϕ^+
$+1$
W^-
Z^0
$\gamma,\ g$
W^+
T_3
-1
0
$+1$
ϕ^-
h^{0*}
-1
-1

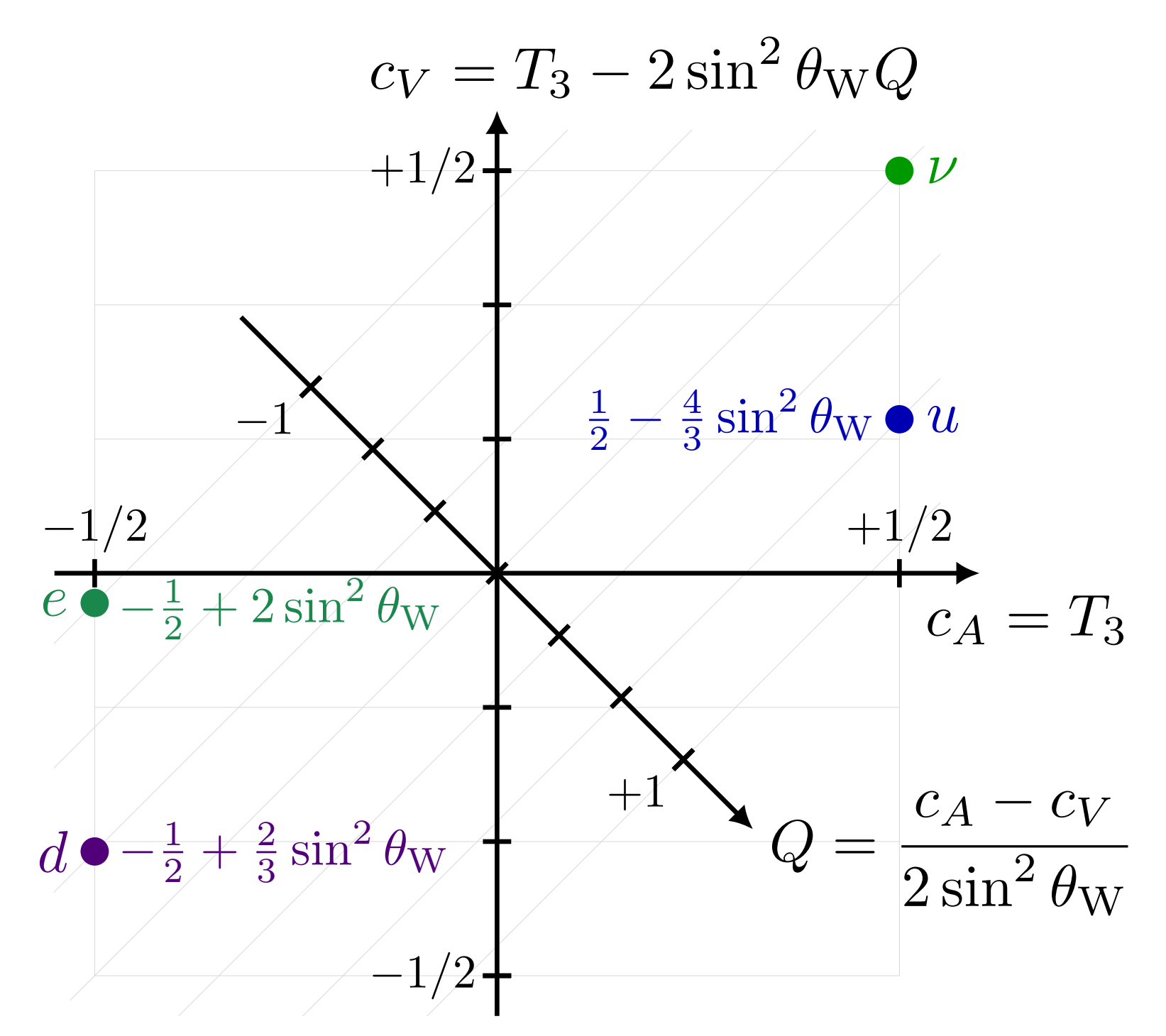

$Y_{\mathrm{W}}/2$
$+1$
$Q = T_3 + \frac{1}{2}Y_{\mathrm{W}}$
ϕ^0
ϕ^+
$+1$
W^-_μ
Z^0_μ
$A^0_\mu,\ G^i_\mu$
W^+_μ
T_3
-1
0
$+1$
ϕ^-
ϕ^{0*}
-1
-1

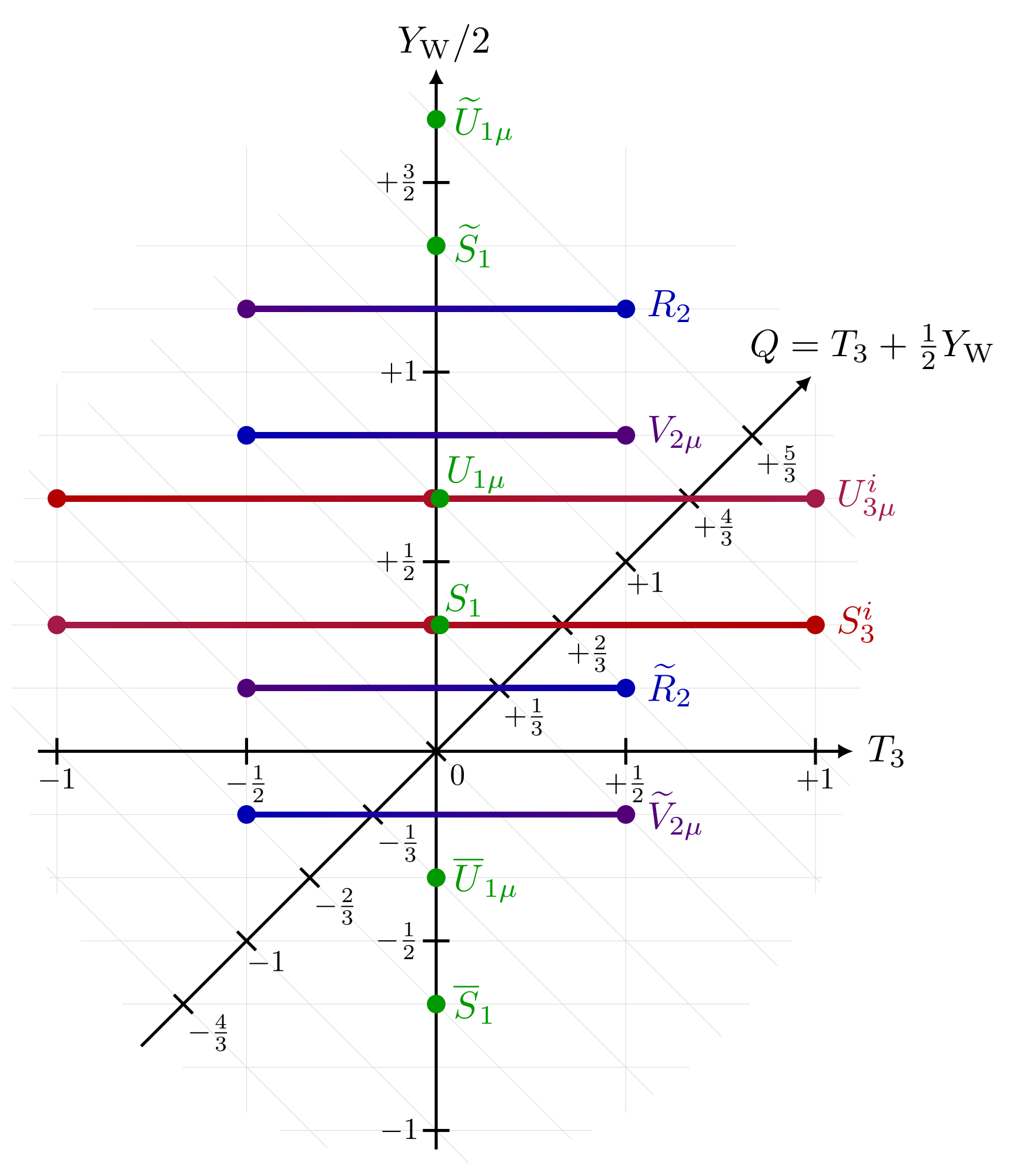

$Y_{\mathrm{W}}/2$
$+1$
$Q = T_3 + \frac{1}{2}Y_{\mathrm{W}}$
ϕ^0
ϕ^+
$+1$
W^-_μ
W^3_μ
B_μ, G^i_μ
W^+_μ
T_3
-1
0
$+1$
ϕ^-
ϕ^{0*}
-1
-1


$Y_W/2$
$+1$
$\overline{e}_R$
$Q = T_3 + \frac{1}{2}Y_W$
u_R
$\overline{\nu}_L$
$\overline{e}_L$
$+1$
$\overline{d}_R$
d_L
u_L
$\nu_R, \overline{\nu}_R$
T_3
-1
0
$+1$
$\overline{u}_L$
$\overline{d}_L$
d_R
e_L
ν_L
-1
$\overline{u}_R$
-1
e_R


$Y_W/2$
$+1$
$\overline{e}_R$
$Q = T_3 + \frac{1}{2}Y_W$
u_R
$\overline{\nu}_L$
$\overline{e}_L$
$\overline{L}_L = (\overline{\nu}, \overline{e})_L$
$+1$
$\overline{d}_R$
d_L
u_L
$Q_L = \begin{pmatrix} u \\ d \end{pmatrix}_L$
$\nu_R, \overline{\nu}_R$
T_3
-1
0
$+1$
$\overline{u}_L$
$\overline{d}_L$
$\overline{Q}_L = (\overline{u}, \overline{d})_L$
d_R
e_L
ν_L
$L_L = \begin{pmatrix} \nu \\ e \end{pmatrix}_L$
-1
$\overline{u}_R$
-1
e_R


$Y_W/2$
+1
+5/6
+2/3
+1/2
+1/3
+1/6
0
−1/6
−1/3
−1/2
−2/3
−5/6
−1
$\overline{e}_R$
u_R
$\overline{\nu}_L$
$\overline{e}_L$
$\overline{d}_R$
d_L
u_L
$\nu_R, \overline{\nu}_R$
$\overline{u}_L$
$\overline{d}_L$
d_R
e_L
ν_L
$\overline{u}_R$
e_R
Q
+1
+2/3
+1/3
0
−1/3
−2/3
−1
−1
0
+1
T_3


$c_V = T_3 - 2\sin^2\theta_\mathrm{W} Q$
$+1/2$
ν
-1
$\frac{1}{2} - \frac{4}{3}\sin^2\theta_\mathrm{W}$
u
$-1/2$
$+1/2$
e
$-\frac{1}{2} + 2\sin^2\theta_\mathrm{W}$
$c_A = T_3$
$+1$
d
$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_\mathrm{W}$
$Q = \dfrac{c_A - c_V}{2\sin^2\theta_\mathrm{W}}$
$-1/2$


$Y_{\rm W}/2$
$\widetilde{U}_{1\mu}$
$+\frac{3}{2}$
$\widetilde{S}_1$
R_2
$+1$
$V_{2\mu}$
$U_{1\mu}$
$U^i_{3\mu}$
$+\frac{1}{2}$
S_1
S^i_3
$\widetilde{R}_2$
-1
$-\frac{1}{2}$
0
$+\frac{1}{2}$
$+1$
T_3
$\widetilde{V}_{2\mu}$
$\overline{U}_{1\mu}$
$-\frac{1}{2}$
$\overline{S}_1$
-1
$Q = T_3 + \frac{1}{2}Y_{\rm W}$
$+\frac{5}{3}$
$+\frac{4}{3}$
$+1$
$+\frac{2}{3}$
$+\frac{1}{3}$
$-\frac{1}{3}$
$-\frac{2}{3}$
-1
$-\frac{4}{3}$